

Project Name: OpenCart E-Commerce Website – Manual Testing

Project Type: Manual Testing Project

Domain: E-Commerce

Testing Type: Manual Testing

Prepared By: Sanika G. Kale

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Project Overview

OpenCart is an open-source e-commerce platform that provides functionality for customers to browse products, search products, manage shopping carts, create accounts, place orders, and perform checkout operations.

The objective of this project is to perform manual testing of the OpenCart e-commerce application to verify that its major functionalities work according to the specified requirements and provide a reliable user experience.

Project Objective

The objective of this project is to perform comprehensive manual testing of the OpenCart e-commerce application by validating both frontend customer functionalities. The project focuses on requirement validation, test case design, functional and non-functional testing, defect identification and reporting, retesting, regression testing, and end-to-end business flow validation to ensure the application meets its functional requirements and provides a reliable user experience.

Project Scope

1. In-Scope

1. **User Registration**
    - New customer registration
    - Mandatory field validation
    - Email validation
    - Password and confirmation password validation
    - Existing email validation
2. **User Login**
    - Valid and invalid credentials
    - Blank fields
    - Password masking
    - Login error messages
3. **Logout**
    - Successful logout
    - Session behavior after logout
    - Accessing protected pages after logout
4. **Forgot Password**
    - Password recovery request
    - Valid/invalid email validation
    - Recovery-related messages
5. **Product Search**
    - Valid product search
    - Invalid product search
    - Partial keyword search
    - Blank search

- Search result validation

6. **Product Display & Details**

   - Product name
   - Product image
   - Price
   - Description
   - Availability
   - Product information

7. **Product Comparison**

   - Add products for comparison
   - Compare product information
   - Remove products from comparison

8. **Wishlist**

   - Add product to wishlist
   - Remove product
   - Wishlist behavior for logged-in/logged-out users

9. **Shopping Cart**

   - Add product
   - Remove product
   - Update quantity
   - Verify price and total
   - Cart persistence

10. **Checkout**

    - Checkout workflow
    - Customer information
    - Billing/shipping address
    - Shipping method
    - Payment method
    - Order confirmation

11. Additional Customer Features

- Product Reviews
- Contact Us
- Newsletter subscription, if available
- Reward Points, if available
- Returns/Return Request, if available

2.Out of Scope

- Source-code/unit testing
- Performance, load, and stress testing
- Security penetration testing
- Server infrastructure testing
- Production infrastructure validation
- Internal testing of third-party payment gateway systems

Testing Type

- Requirement Analysis and Test Planning
- Test Scenario and Test Case Design
- Test Data Preparation and Test Execution
- Functional, UI, Usability, Positive, and Negative Testing
- Smoke, Sanity, Retesting, Regression, Exploratory, and End-to-End Testing
- Compatibility testing across selected supported browsers
- Defect Reporting and Test Closure

Test Design Techniques

Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, State Transition Testing, Error Guessing, and Use Case Testing are used to design effective test cases for the OpenCart application. These techniques help ensure adequate coverage of valid, invalid, boundary, business-rule, state-based, and end-to-end scenarios across both the customer frontend and admin backend.

Test Environment

Parameter	Details
Application	OpenCart E-Commerce Application
Testing Approach	Manual Testing
Operating System	Windows
Browsers	Google Chrome / Microsoft Edge as applicable
Test Data	Valid and invalid customer/admin/product data
Documentation	Microsoft Excel / PDF
Defect Tracking	Jira or structured defect spreadsheet, as applicable

Defect Management

Defects identified during execution will be documented with a unique defect ID, title, module, environment, preconditions, reproduction steps, expected result, actual result, severity, priority, evidence, and status.

Defect lifecycle: New > Assigned > Fixed > Retest > Closed / Reopened.

Entry Criteria

Requirements are available and understood, the OpenCart application is deployed and accessible, the test environment is ready, required frontend/admin credentials and test data are available, and test scenarios and test cases are prepared for execution.

Exist Criteria

All planned test cases are executed, critical functionality is validated, identified defects are documented and tracked, required retesting and regression testing are completed, test results are documented, RTM is updated, and the final Test Summary Report is prepared.

Roles and Responsibilities

Responsible for analyzing requirements, preparing test scenarios and test cases, executing functional and non-functional test activities, identifying and reporting defects, performing retesting and regression testing, validating customer functionalities, maintaining testing documentation, and preparing the final Test Summary Report.

Risk and Assumptions

Risks: Application downtime, unstable test environment, unavailable test data, requirement changes, application blockers, browser compatibility issues, third-party dependencies, and limited testing time may affect test execution and coverage.

Assumptions: The OpenCart application, test environment, requirements, required credentials, test data, and supported browsers are available. It is assumed that identified defects can be tracked and fixed builds will be available for retesting when required.

Conclusion

The OpenCart Manual Testing project demonstrates a structured and systematic approach to testing an e-commerce application. The project covers both the customer-facing frontend panel, including user registration, login, product search, shopping cart, checkout, and other relevant functionalities.

The testing process includes requirement analysis, test planning, test scenario and test case design, test execution, defect identification and reporting, retesting, regression testing, and test closure. Test design techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, State Transition Testing, Error Guessing, and Use Case Testing are used to achieve effective test coverage.

The project also maintains professional QA documentation such as FRS, Test Plan, Test Scenarios, Test Cases, RTM, Test Execution Report, Defect Reports, Screenshots, and Test Summary Report.